

PUSECGIS — MVP Specification

Status: Draft 0.1 — Proposal | **Date:** 2026-09-02

Scope: Barnes-Jewish Hospital Public Safety — Common Operating Picture

Goal: A shared visual map capturing incidents, problem areas, and shift-to-shift information across BJC's campus — broadcast to all satellite offices and ED desks.

1. Concept

What it does: A shared operating map for BJC Public Safety that captures incidents (manual entry from officers/dispatch), visualizes them geographically on campus floor plans, and broadcasts live status to all satellite offices and ED desks.

Why it matters: Currently, shift-to-shift information is lost over radio. Officers walk a floor, note problem areas, agitated patients, duress events — then that knowledge evaporates when shift changes. This tool preserves that information visually.

Who it's for: BJC Public Safety officers, floor rounding staff, Dispatch, and Public Safety leads.

Where it runs: BJC internal network only. No internet dependency.

2. Display & Users

Broadcast Screens (6 total — read-only, auto-refreshing)

Main Public Safety Control	TV/Monitor	Primary situational awareness hub
Satellite Office #1	TV/Monitor	Regional status visibility
Satellite Office #2	TV/Monitor	Regional status visibility
Satellite Office #3	TV/Monitor	Regional status visibility
Childrens' ED Desk	Monitor	Pediatric ED staff awareness
Adult ED Desk	Monitor	Adult ED staff awareness

Officer Workstations (write-access)

Floor Rounding Officers	Log incidents, add notes
Dispatch	Log incidents, add notes
Public Safety Leads	Edit/close incidents, view reports
Admin	Manage users, locations, config

3. Data Model

Incident Types & Color Map:

Victim of Violence	Red	#DC2626	Assault, physical altercations
Agitated Visitor	Orange	#EA580C	Family/visitor agitation
Problematic Patient	Yellow	#D97706	Behavioral, recurring disturbance
Patient with Sitter	Blue	#2563EB	Room sitter assigned
Elopment Patient	Green	#16A34A	Wandering attempt, elopment alert
Duress Alarm Call	Dark Gray	#374151	Staff duress button triggered
Hardware / Facility Issue	Gray	#6B7280	Broken door, camera, lock, alarm
General Safety Concern	Purple	#7E22CE	Suspicious activity, problem area notes

Core Tables

shifts: id, shift_date, shift_code (DAY/EVE/NIGHT), start_time, end_time, created_at

incidents: id, shift_id, incident_type, location_ref (text), description, status (open/resolved/monitoring), logged_by, created_at, updated_at

shift_handoff: id, shift_id, location_ref (nullable), note, logged_by, created_at

users: id, username, display_name, role (officer/dispatch/lead/admin), active

4. API Endpoints

Auth: POST /api/auth/login, GET /api/auth/me

Incidents (auth required): GET /api/incidents?status=&type;=&date;=, POST /api/incidents, PUT /api/incidents/:id, DELETE /api/incidents/:id

Broadcast (no auth): GET /api/broadcast/incidents?hours=2, GET /api/broadcast/config

Handoff: GET /api/handoff/notes?shift=&date;=, POST /api/handoff/notes

Locations/Users: GET /api/locations?q=, GET /api/users

5. UI Screens

Officer UI: Map view with incident plotting, quick log form (Type dropdown, Location text/search, Description text), Open/Resolved toggle, Shift handoff note area, Incident list with filters.

Broadcast UI (Kiosk): Fullscreen map with color-coded markers, auto-refresh every 30 seconds, incident sidebar (scrollable, newest at top), shift indicator, filter bar, kiosk-safe (fullscreen, prevents exit, auto-reloads).

6. Tech Stack

Backend	Python 3.11, FastAPI, Uvicorn	Clean, fast, extensible
Database	PostgreSQL 16 + PostGIS 3.4	Mature open-source GIS
Frontend (Officer)	React 18, Vite, Leaflet	Component framework + mapping
Frontend (Broadcast)	React 18, Leaflet, Kiosk wrapper	Reuses map engine, read-only
Auth	Session cookies	Internal only, no SSO for MVP
Proxy	Nginx	Routing, static files, HTTPS
Hosting	Linux server, BJC internal	VM or bare metal
Backup	Daily pg_dump	Simple, reliable

7. Implementation Phases

Phase 1: MVP Core (Weeks 1-3)

- PostgreSQL + PostGIS setup
- FastAPI backend (Incidents, Shifts, Users, Handoff, Auth)
- Officer UI (map, incident entry, filter, handoff)
- Broadcast UI (fullscreen map, sidebar, 30s auto-refresh)
- Kiosk mode script (auto-fullscreen, auto-restart)
- Deploy to BJC internal network
- Smoke test with 2-3 officers

Phase 2: Enhancements (Weeks 4-6)

- Duress alarm CSV import & parser
- Hotspot heatmap overlay
- Floor plan overlay (PDF base map)
- Trend charts / reports (weekly/monthly)
- Exportable reports (PDF/CSV)

Phase 3: Integration (Weeks 7-10)

- Duress alarm system API integration
- Radio/CAD integration (if available)
- Mobile-responsive officer UI
- RBAC (officer vs. lead vs. admin)
- Audit logging & security hardening

8. Acceptance Criteria (MVP)

- Officers can log incidents (type, location, notes)
- Incidents appear on map in correct colors
- Broadcast screens show all active incidents with 30s auto-refresh
- Broadcast screens display at 1920x1080 without map scrolling
- Shift handoff notes are captured and visible per shift
- 6 display targets can view broadcast simultaneously
- All runs on BJC internal network (no internet dependency)
- CSV duress alarm import works (Phase 2)

9. Open Questions

1. Floor plan format — PDF exists, are CAD files available?
2. MTF CSV sample — can we get one export?
3. User roster — how many active officers/dispatch?
4. Server specs — VM or bare metal? RAM, storage?
5. Duress alarm system vendor — Cognosos, kontakt.io, etc.? API docs?
6. Severity levels — need 1-5 rating or type is enough?
7. Incident retention — how long should resolved incidents stay visible?
8. Offline mode — should broadcast cache if network drops?
9. Notifications — audible/visual alerts for new incidents?
10. Escalation workflow — auto-escalate persistent incidents?